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Programme: T. Y. B.Sc. IT

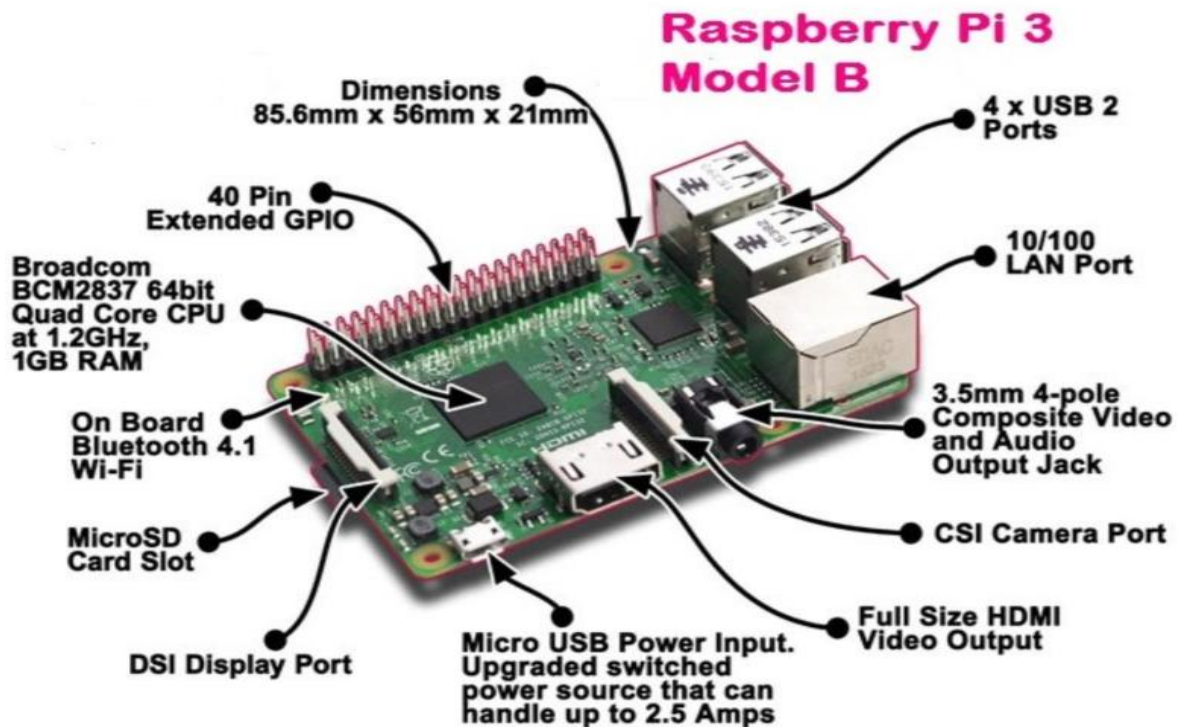
Semester: 6

Course: Internet of Things

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Practical 1

Aim: Starting Raspbian OS, Familiarizing with Raspberry Pi Components and interface, Connecting to ethernet, Monitor, USB.



This image shows the parts of the Raspberry Pi 3 Model B

1. GPIO (General Purpose Input/Output) Pins

- These are small pins on the board.
- They are used to connect electronic components like LED, sensors, motors, and buttons.
- They help the Raspberry Pi send and receive signals.

2. DSI Display Port

- DSI means Display Serial Interface.
- It is used to connect the official Raspberry Pi display screen.

3. CSI Camera Port:

- CSI means Camera Serial Interface.
- It is used to connect the Raspberry Pi camera module for taking photos and videos.

4. **MicroSD Slot:**

- This slot is used to insert a MicroSD card.
- The operating system and all files are stored on this card.

5. **HDMI / USB / Network Ports:**

- **HDMI Port:** Used to connect a monitor or TV for display.
- **USB Ports:** Used to connect keyboard, mouse, pendrive, etc.
- **Network (LAN) Port:** Used to connect internet using a LAN cable.

6. **Micro USB Power Port:**

- This port is used to give power to the Raspberry Pi.
- Without power, the Raspberry Pi will not start.

Practical 2

Aim: Displaying different LED patterns with Raspberry Pi.

a) Single LED brightness.

```
import time

import RPi.GPIO as GPIO

GPIO.setmode(GPIO.BOARD)

GPIO.setwarnings(False)

LED = 11

GPIO.setup(LED, GPIO.OUT)

while True:

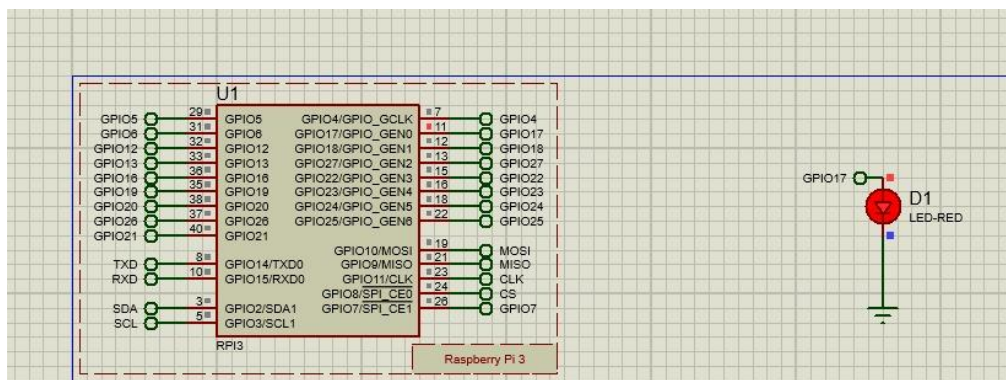
    GPIO.output(LED, True)

    time.sleep(3)

    GPIO.output(LED, False)

    time.sleep(3)
```

OUTPUT:



b) Alternate Blinking LED Pattern:

```
import RPi.GPIO as GPIO

import time

GPIO.setmode(GPIO.BCM)

led1 = 17

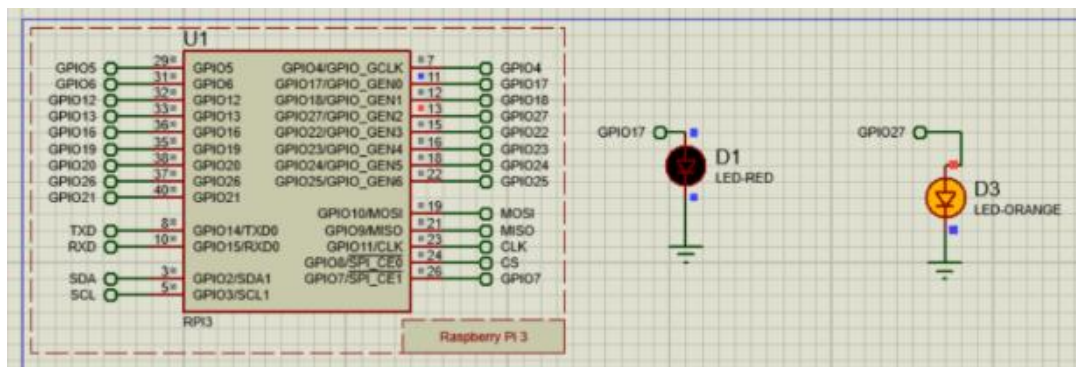
led2 = 27
```

```

GPIO.setup(led1, GPIO.OUT)
GPIO.setup(led2, GPIO.OUT)
while True:
    GPIO.output(led1, True)
    GPIO.output(led2, False)
    time.sleep(0.3)
    GPIO.output(led1, False)
    GPIO.output(led2, True)
    time.sleep(0.3)

```

OUTPUT:



C) Build Traffic signal circuit:

```

import RPi.GPIO as GPIO
import time

GPIO.setmode(GPIO.BCM)

RED = 17
YELLOW = 27
GREEN = 22

GPIO.setup(RED, GPIO.OUT)
GPIO.setup(YELLOW, GPIO.OUT)
GPIO.setup(GREEN, GPIO.OUT)

try:
    while True:

```

```
# RED ON
```

```
GPIO.output(RED, GPIO.HIGH)
```

```
GPIO.output(YELLOW, GPIO.LOW)
```

```
GPIO.output(GREEN, GPIO.LOW)
```

```
print("STOP")
```

```
time.sleep(5)
```

```
# YELLOW ON
```

```
GPIO.output(RED, GPIO.LOW)
```

```
GPIO.output(YELLOW, GPIO.HIGH)
```

```
GPIO.output(GREEN, GPIO.LOW)
```

```
print("READY")
```

```
time.sleep(2)
```

```
# GREEN ON
```

```
GPIO.output(RED, GPIO.LOW)
```

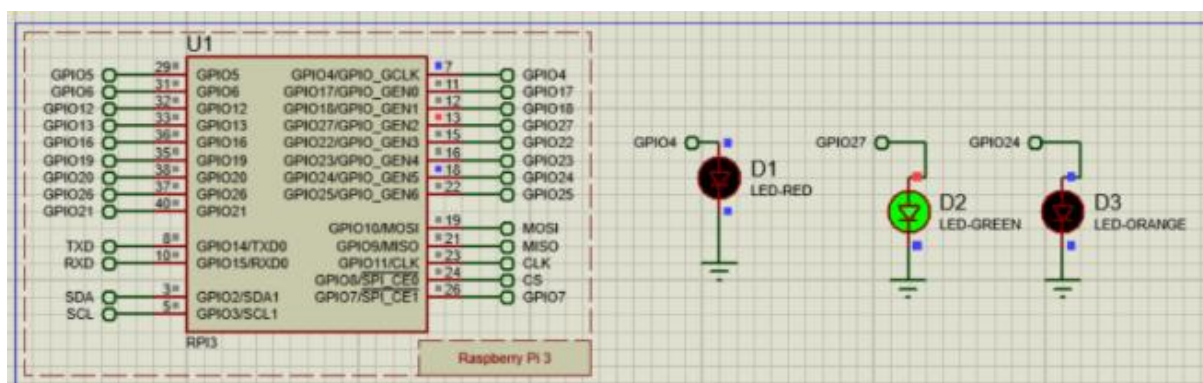
```
GPIO.output(YELLOW, GPIO.LOW)
```

```
GPIO.output(GREEN, GPIO.HIGH)
```

```
print("GO")
```

```
time.sleep(5)
```

OUTPUT:



Practical 3

Aim: Control LED brightness using PWM.

```
import RPi.GPIO as GPIO

import time

GPIO.setmode(GPIO.BCM)

LED = 18

GPIO.setup(LED, GPIO.OUT)

pwm = GPIO.PWM(LED, 1000)

pwm.start(0)

try:

    while True:

        for duty in range(0, 101): # 0–100%

            pwm.ChangeDutyCycle(duty)

            time.sleep(0.02)

        for duty in range(100, -1):

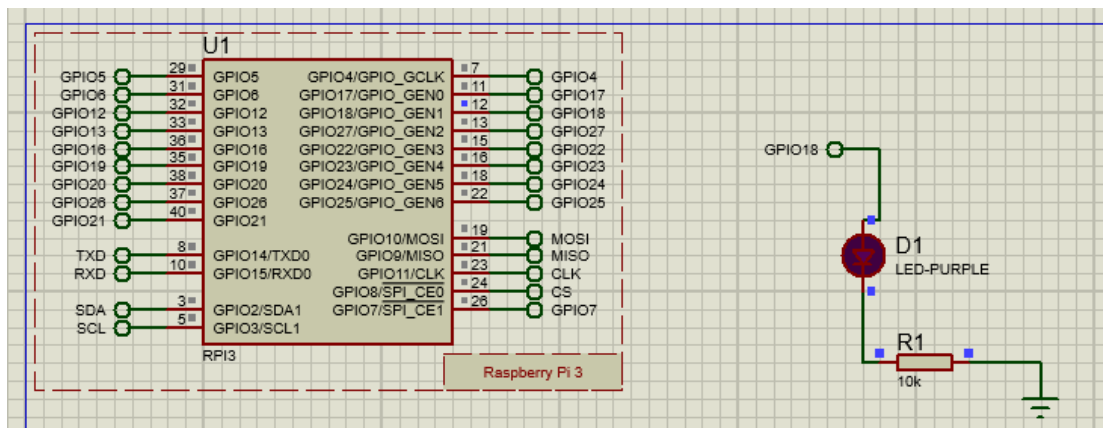
            pwm.ChangeDutyCycle(duty)

            time.sleep(0.02)

except:

    print("error")
```

OUTPUT:



Practical 4

Aim: Control Stepper motor using Raspberry Pi.

```
import RPi.GPIO as GPIO
from time import sleep

GPIO.setmode(GPIO.BOARD)

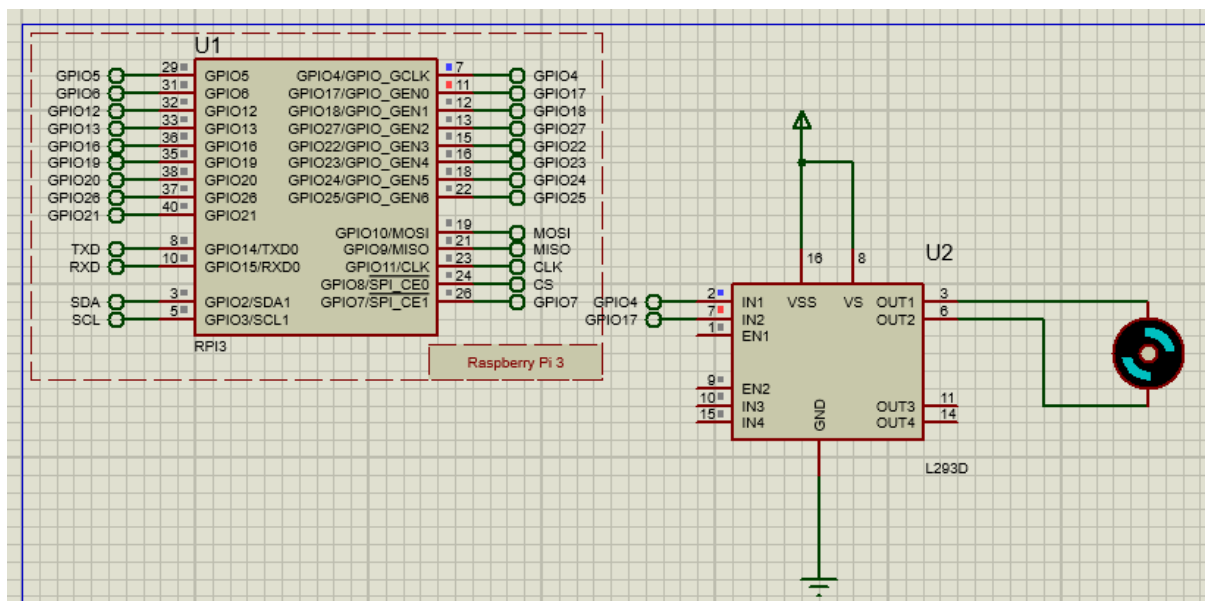
motor1=7
motor2=11

GPIO.setup(motor1,GPIO.OUT)
GPIO.setup(motor2,GPIO.OUT)

while True:

    GPIO.output(motor1,GPIO.HIGH)
    GPIO.output(motor2,GPIO.LOW)
    sleep(3)
    GPIO.output(motor1,GPIO.LOW)
    GPIO.output(motor2,GPIO.HIGH)
    sleep(3)
```

OUTPUT:

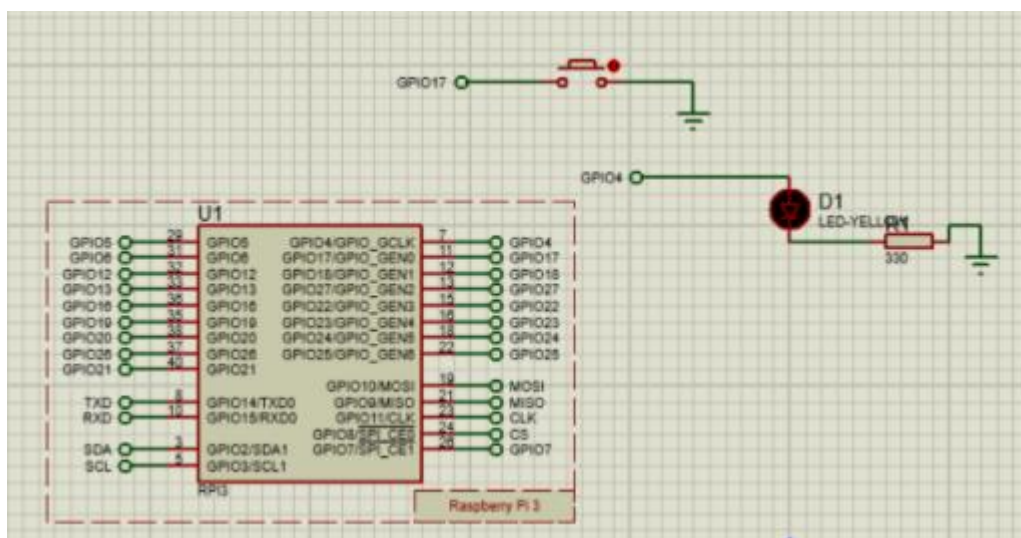


Practical 5

Aim: Control LED using Push Button.

```
import RPi.GPIO as GPIO
import time
GPIO.setmode(GPIO.BOARD)
BUTTON_PIN = 11
LED_PIN = 7
GPIO.setup(BUTTON_PIN, GPIO.IN, pull_up_down=GPIO.PUD_UP)
# Set pin 11 as input with pull-up resistor
GPIO.setup(LED_PIN, GPIO.OUT)
while True:
    if GPIO.input(BUTTON_PIN) == GPIO.LOW:
        # Check if button is pressed (LOW because of pull-up)
        GPIO.output(LED_PIN, GPIO.HIGH)
    else:
        GPIO.output(LED_PIN, GPIO.LOW)
    time.sleep(0.1)
```

OUTPUT:



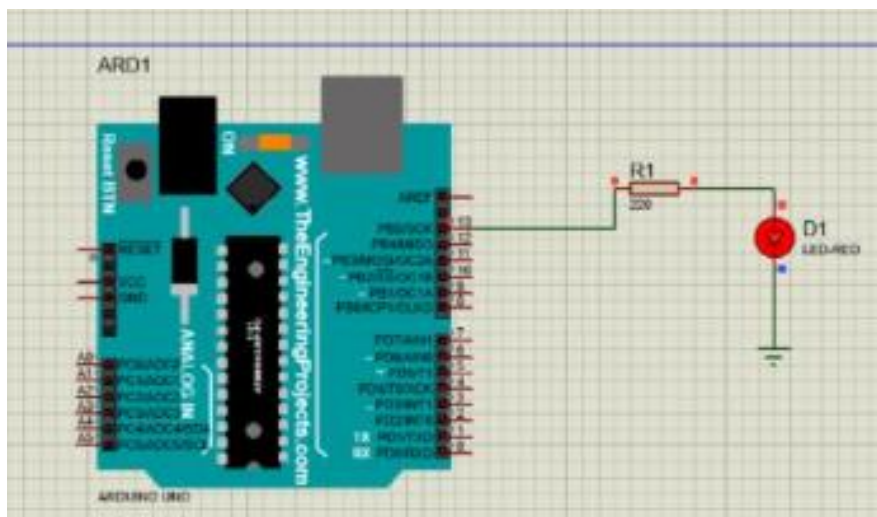
Practical 6

Aim: Displaying different LED patterns with Arduino Uno.

A) Single LED:

```
void setup() {  
  pinMode(13, OUTPUT); // Set pin 13 as output  
}  
  
void loop() {  
  digitalWrite(13, HIGH); // Turn LED on  
}
```

OUTPUT:

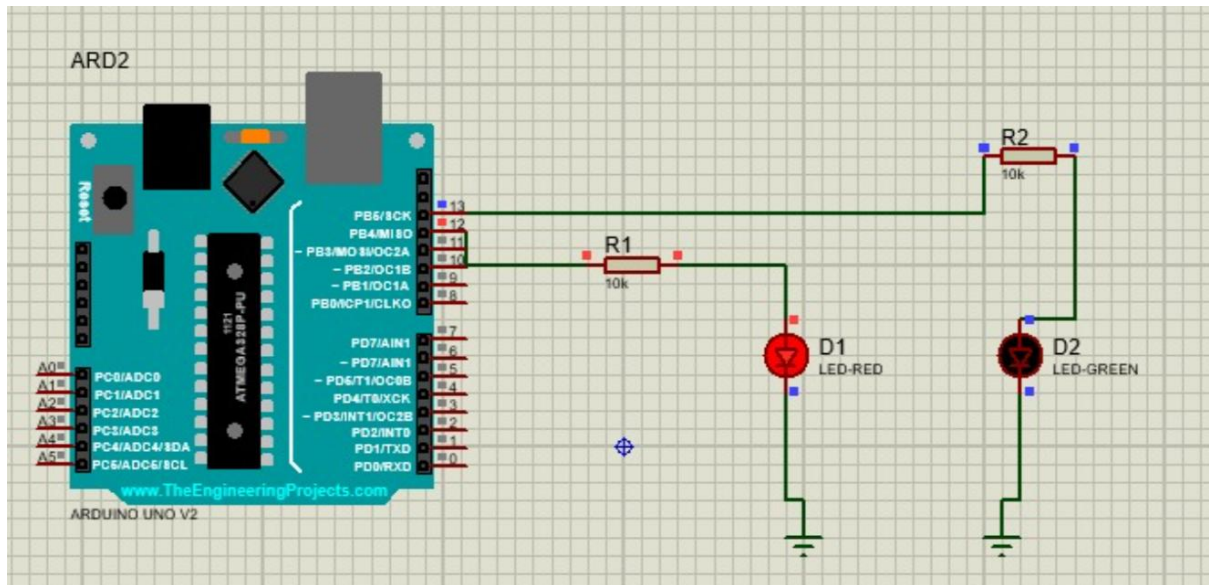


B) ON/OFF LED:

```
void setup() {  
  pinMode(13, OUTPUT);  
}  
  
loop() {  
  digitalWrite(13, HIGH);  
  delay(1000);  
  digitalWrite(13, LOW);  
  delay(1000);  
}
```

}

OUTPUT:



C) Traffic Signal:

```
int red = 10;

int yellow = 9;

int green = 8;

void setup() {
  pinMode(red, OUTPUT);
  pinMode(yellow, OUTPUT);
  pinMode(green, OUTPUT);
}

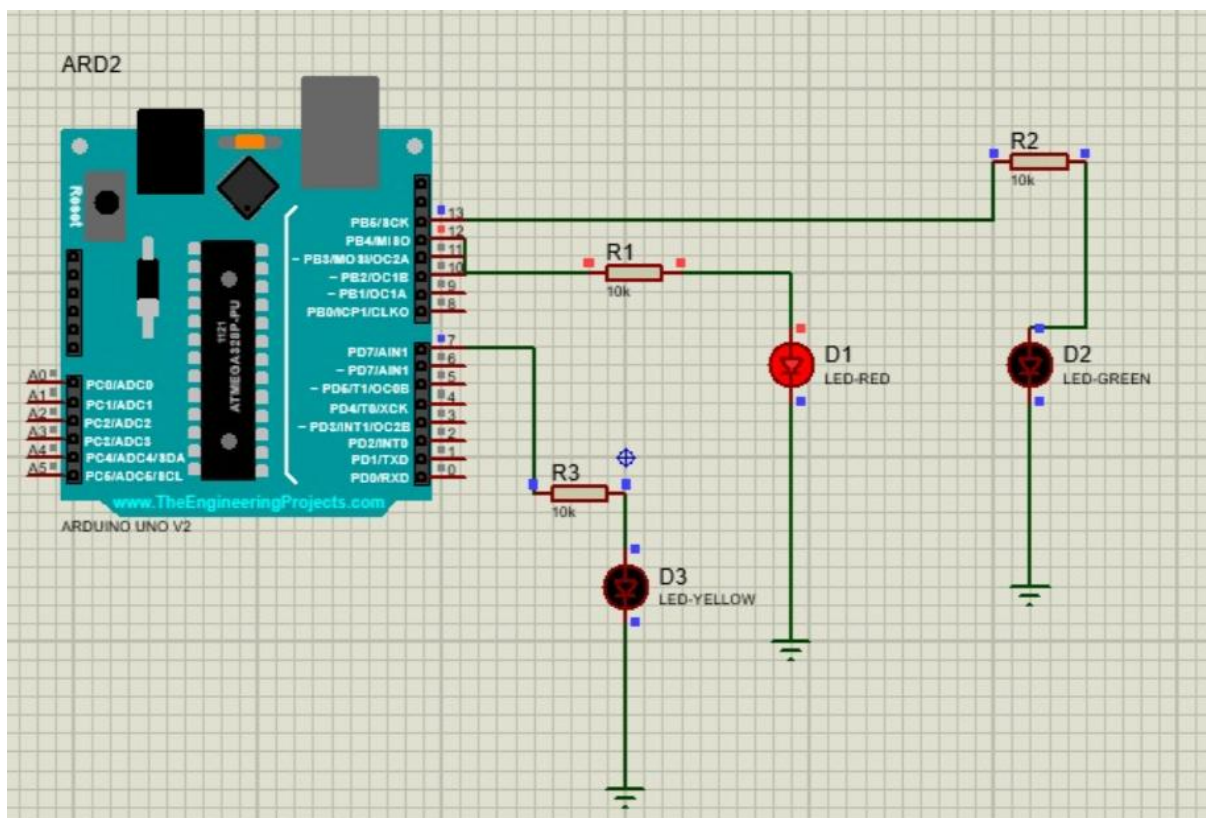
void loop() {
  digitalWrite(green, HIGH);
  digitalWrite(yellow, LOW);
  digitalWrite(red, LOW);
  delay(3000);
  digitalWrite(green, LOW);
  digitalWrite(yellow, HIGH);
```

```

digitalWrite(red, LOW);
delay(2000);
digitalWrite(green, LOW);
digitalWrite(yellow, LOW);
digitalWrite(red, HIGH);
delay(5000);
}

```

OUTPUT:

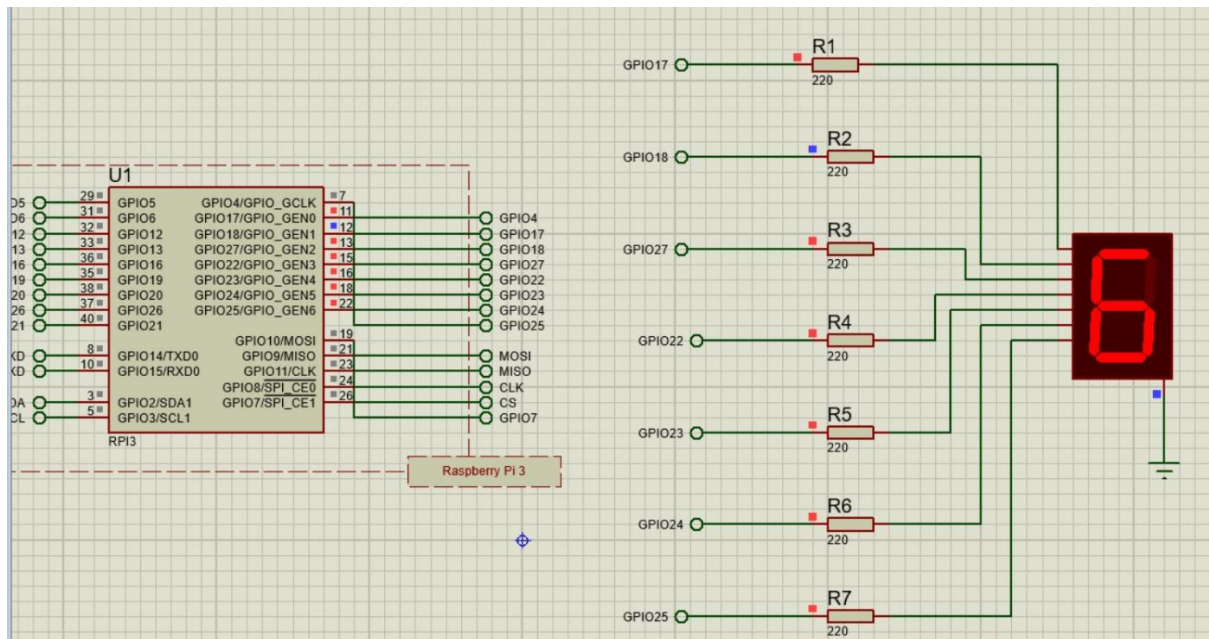


Practical 7

Aim: Displaying digit over 7-Segment Display using Raspberry Pi.

```
import RPi.GPIO as GPIO
import time
GPIO.setmode(GPIO.BCM)
segments = [17,18,27,22,23,24,25]
for pin in segments:
    GPIO.setup(pin, GPIO.OUT)
    GPIO.output(pin, 0)
digits = [
    [1,1,1,1,1,1,0], # 0
    [0,1,1,0,0,0,0], # 1
    [1,1,0,1,1,0,1], # 2
    [1,1,1,1,0,0,1], # 3
    [0,1,1,0,0,1,1], # 4
    [1,0,1,1,0,1,1], # 5
    [1,0,1,1,1,1,1], # 6
    [1,1,1,0,0,0,0], # 7
    [1,1,1,1,1,1,1], # 8
    [1,1,1,1,0,1,1] # 9
]
while True:
    for num in digits:
        for i in range(7):
            GPIO.output(segments[i], num[i])
        time.sleep(1)
```

OUTPUT:



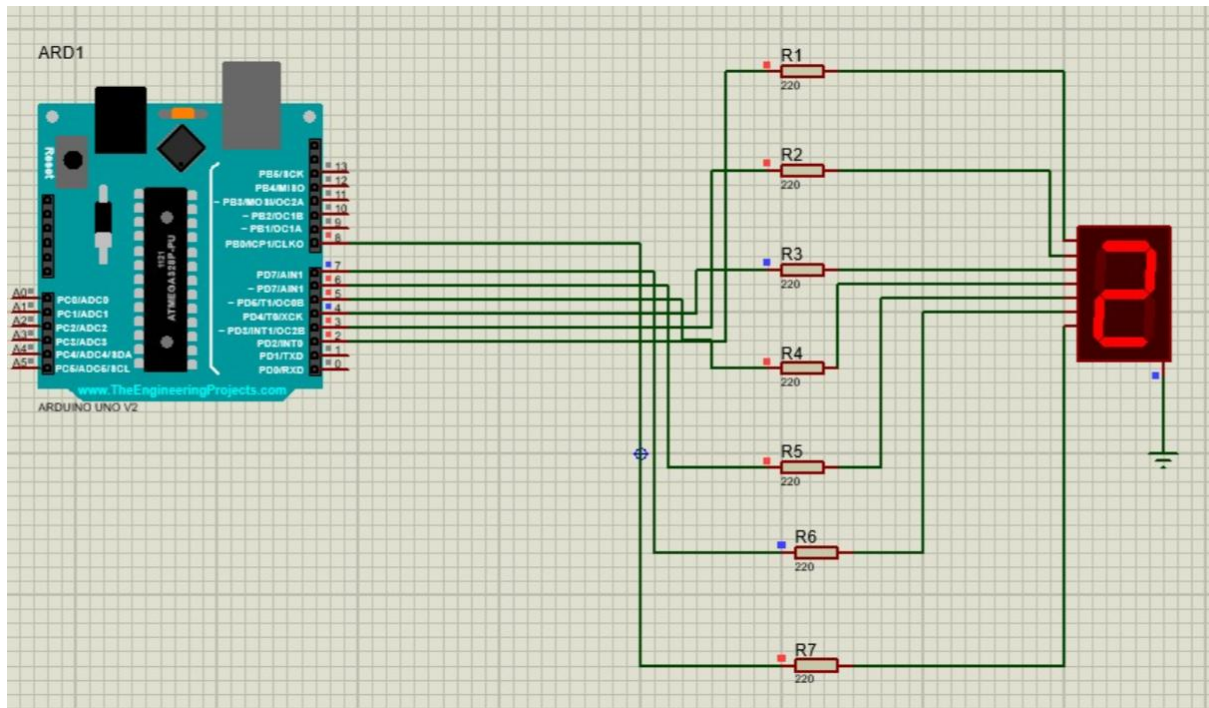
Practical 8

Aim: Displaying digit over 7-Segment Display using Arduino.

```
int segPins[7] = {2,3,4,5,6,7,8};  
byte digits[10][7] = {  
  {1,1,1,1,1,1,0}, //0  
  {0,1,1,0,0,0,0}, //1  
  {1,1,0,1,1,0,1}, //2  
  {1,1,1,1,0,0,1}, //3  
  {0,1,1,0,0,1,1}, //4  
  {1,0,1,1,0,1,1}, //5  
  {1,0,1,1,1,1,1}, //6  
  {1,1,1,0,0,0,0}, //7  
  {1,1,1,1,1,1,1}, //8  
  {1,1,1,1,0,1,1} //9  
};  
  
void setup() {  
  for(int i=0;i<7;i++) {  
    pinMode(segPins[i], OUTPUT);  
  }  
}  
  
void loop() {  
  for(int num=0; num<10; num++) {  
    for(int i=0;i<7;i++) {  
      digitalWrite(segPins[i], digits[num][i]);  
    }  
    delay(1000);  
  }  
}
```

}

OUTPUT:



Practical 9

Aim: Control LED using Push Button Arduino.

```
const int ledPin = 13;  
  
const int buttonPin = 2;  
  
void setup() {  
  pinMode(ledPin, OUTPUT);  
  pinMode(buttonPin, INPUT_PULLUP);  
}  
  
void loop() {  
  int buttonState = digitalRead(buttonPin);  
  
  if(buttonState == LOW) {  
    digitalWrite(ledPin, HIGH);  
  }  
  
  else {  
    digitalWrite(ledPin, LOW);  
  }  
}
```

OUTPUT:

